

Event Management Portal - Architecture & Technical Documentation

This document describes the recommended architecture, technology stack, workflows, and cost estimations for building an Event Management Portal for doctor registration and event attendance.

1. Project Overview

- Doctors register for events using mobile number, doctor ID, name, and email.
- Events can be free or paid.
- Paid events require payment gateway integration.
- Confirmation email with QR code is sent 2–3 days before event.
- QR code is scanned at event venue for check-in and badge generation.

2. Recommended Technology Stack

Frontend:

- Angular or React
- Tailwind CSS / Bootstrap

Backend:

- Node.js + Express.js

Database:

- MongoDB or PostgreSQL

Authentication:

- JWT / OTP login via mobile

Payment Gateway:

- Razorpay (recommended for India) or Stripe

Email Service:

- AWS SES / SendGrid

QR Code:

- QR code generator library + scanner integration

Deployment:

- Frontend: Vercel / Netlify
- Backend: AWS / Render / Azure

3. Architecture Flow

User → Frontend Portal → Backend API → Database

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Payment Gateway

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Email Service (QR)

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Event Check-In Scanner

4. Registration Workflow

1. Doctor opens registration portal.
2. Selects event.
3. Enters details.
4. Completes payment if event is paid.
5. Registration saved in database.
6. Confirmation email sent.

5. QR Workflow

- Before event, system generates unique QR code.
- Email sent to participant.
- At event entry, QR scanned.
- Status updated to Checked-In.
- Badge/tag printed.

6. Cost Estimation

Example:

1000 users × ■500 ticket = ■5,00,000 revenue

Payment Gateway Fees:

Razorpay/Stripe (approx 2% domestic)

2% of ₹5,00,000 = ₹10,000 gateway fee

Net amount after gateway fee:

₹4,90,000

Email Cost:

AWS SES usually very low (<₹100–₹500 for this scale)

7. Recommended Features

- Admin dashboard
- Event analytics
- Check-in dashboard
- Badge printing
- Email reminders
- Export attendee reports

8. Recommendation

For this project, Angular + Node.js + MongoDB + Razorpay is a strong and scalable stack for India-based operations.